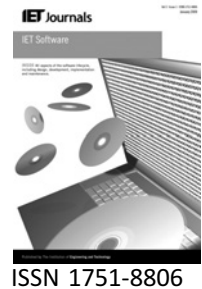


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Knowledge repository to improve agile development processes learning

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Abstract: Software process learning depends significantly on technologies to store, search and share knowledge that can be used in software projects. The aim of this work is to provide a set of guidelines to develop a knowledge-based process asset libraries (PALs), using a Wiki (as a Web 2.0 technology), to store software engineering best practices. Fieldwork was carried out in a two-year training course of agile development process. This was validated in two phases (with and without PAL), each of which consisted of two stages: training and project. The study demonstrates that on the one hand, the learning process can be improved using a PAL–Wiki to transfer and share the software process knowledge, and on the other hand, junior software engineers developed software products with a greater degree of independence.

1 Introduction

According to [1], human factors represent one of the most important areas in software engineering improvement. Therefore for software engineers, continuous learning is a cornerstone for both academia and industry. This paper focuses on the use of Wikis as Web 2.0 knowledge repositories to help software engineers to access, query and reuse process assets in projects, improving in consequence the learnability and ease of use of software engineering best practices.

A software process is a coherent set of policies, organisational structures, technologies and artefacts that are required to design, develop, install and maintain software products [2]. Software engineers are not always motivated to follow the process if they lack supporting technology to create, store and search knowledge needed in order to develop their activities [3, 4]. Moreover, increasing employee turnover rates and a more mobile workforce increase the need for knowledge about processes and projects to be stored and reused [5]. Time and effort to create a project process description are reduced when a reusable process is instantiated instead of building the process from scratch [6]. Organisations need technological infrastructures to manage software process knowledge so that users can learn, use and adapt the process [7] and to carry out projects using repositories that foster collaboration.

One such tool is a process asset library (PAL). PAL is a knowledge repository used to keep and make available all process assets that are useful to those who are defining, implementing and managing processes in an organisation [8]. However, for PAL to be effective, it is necessary to develop domain-specific process assets that can be easily stored and updated [9]. Moreover, PAL is required to capture the diverse forms of knowledge gained from projects. PAL must store useful and essential artefacts to help create the work products of the projects. Also, it must provide learning capabilities to develop collective skills and be an effective learning environment for training new employees [10].

The objective is for software engineers to have greater autonomy in learning; there is greater emphasis on active learning, with creation and participation playing key roles [11]. During training, software engineers can be involved in a few small software development projects that must be accomplished quickly. At the same time, they must obtain theoretical knowledge about software processes. Therefore tools are required to provide different support levels for learning and using processes [4]. Although the number of software processes knowledge repositories is limited, in agile development it is even more limited, mainly to file repositories. In this regard, the challenge is two-fold: first, we defined the knowledge management mechanisms that allow a PAL to improve learning and using the software

process. Then, we applied the theoretical guidance defined for agile development processes repositories.

None of these can be developed without the implementation of appropriate mechanisms using the necessary technology. Jones [12] has shown that Wikis are among the key technologies to implement software engineering best practices for learning new skills. Wikis are well positioned in the ranking of software learning channels in terms of cost, efficiency, effectiveness and currency. Therefore Wikis are ideal technologies for implementing PALs.

To meet these challenges, we defined a PAL, based on knowledge management, implemented through a Wiki in order to:

- improve agile development processes learning by storing and reusing the process assets and
- allow software engineers to use processes learned with a greater degree of independence.

To address the above objectives, this study proposes using knowledge management. Knowledge management targets the capture, codification and dissemination of knowledge across organisation to enhance value [13]. In software engineering, there has been much discussion about how to manage knowledge [14]. To facilitate the development of knowledge management systems, Web 2.0 offers a more interactive and collaborative framework of technologies [15–17]. The software industry is ideally placed to take advantage of social media technologies, tools and environments [18, 19]. Knowledge management in PALs will increase skills to develop and store knowledge of organisational processes. Web 2.0 includes the concepts of weblogs, Wikis, social tagging, mashups and user-defined content. As a vehicle for implementing the solution proposed, the concept of Wiki is used. A PAL can be implemented by a Wiki to behave like a lightweight repository of knowledge in order to store, organise, distribute and preserve process assets, and to generate shared knowledge.

A Wiki-based PAL to support agile software process learning and use was validated in a training course.

This paper is organised as follows. Section 2 presents related work with PAL. Section 3 describes the proposed solution. Next, Section 4 describes the validation process. Then, Section 5 analyses and discusses the results. Finally, in Section 6, the authors present their conclusions.

2 Related work

In this section, we review some proposals oriented to achieving the goals listed above.

On the web, there are some specific repositories that implement corporate PALs. The Spawar System Center

Pacific's PAL [20] and Goddard Space Flight Center of NASA's PAL [21] include artefacts such as guidelines, policies, standards, procedures and training material. These PALs have a large amount of information that makes it difficult to understand the processes. They are organised by areas or categories but the assets are dispersed, so it is necessary to represent the evolution of assets and, to include easy and flexible editing capabilities.

This work is related to the works of several authors [22], who recommend knowledge management strategies and the use of Wikis to create and maintain process information collaboratively and dynamically.

In the literature, the Wikis are used to support activities such as requirements and specifications, architectural models and test cases. Software development teams have begun using Wikis in processes such as project management and tracking [23]; communication and collaboration; technical documentation of software; architectural design [24]; and management of lessons learned [25].

Software process learning is generally team-based and project-centred [26] so that software engineers have a sound technical knowledge. Today, there is an ever-increasing range of e-learning technologies available for the more traditional teaching and learning strategies; Web 2.0 technologies such as social networking tools, learning portals, learning object repositories, blogs and Wikis [27, 28]. Wikis help to learn the processes to make things easy and understandable and teach engineers how to construct the knowledge and solve the problems they face. Common pedagogical applications of Wikis are supporting writing instruction [29], project-based learning [30] and online/distance education.

There are a few studies on the use of Wikis as PALs. Wongboonsin [31] developed a PAL using a Wiki. The process assets used in their study are based on the capability maturity model integration model. Our study differs in the process structure, the types of artefacts used and the experiment conducted during software practices and projects.

In general, the related work lack functionalities to address some relevant aspects of PALs such as assistants to deliver information about processes and to support the interpretation and application context [32]; an integrated approach to codification (to store information that constitutes the knowledge of the company) and personalisation (to support the flow of information in a company by having a centralised store of information about knowledge sources) [14]; and support to reuse non-structured elements [33].

The solutions/studies mentioned should include knowledge management capabilities support to manage process assets (acquisition, organisation, distribution, use and preservation

of knowledge) [22]. In activities such as knowledge acquisition and organisation, these proposals store formal structured and standardised knowledge according to a reference model. On the other hand, the informal knowledge about the process is generally not included. In terms of knowledge distribution, search of assets are provided, but with very simple formats. Regarding the usage and reuse of knowledge, the tools do not include the ability to instantiate assets to be used in specific software projects. Finally, the evolution and preservation of knowledge should be done collaboratively in PAL.

The main handicap of the works mentioned is that the PALs implemented are no longer static repositories and must become useful and active knowledge repositories of software processes. Systems based on Web 2.0 such as Wikis offer advantages for developing and implementing lightweight knowledge repositories. The processes of traditional knowledge management such as creation, transfer, storage and retrieval are supported by Wiki systems [34]. Wikis are easy-to-use knowledge-sharing tools and can be oriented to helping development teams collaborate and work together to carry out activities and manage software projects through a centralised repository of artefacts [4].

3 Solution proposed

This paper seeks to apply knowledge management concepts to process assets through a PAL-Wiki to improve learning and use of software processes. The proposal offers an integrated approach to knowledge management strategies, based on codification and personalisation to store artefacts, and allows users to share knowledge. It also incorporates knowledge management activities in order to acquire, organise, distribute, use, preserve, manage and measure the assets. The guidelines for building a knowledge-management-based PAL include the definition of a structural model and an incremental processes model. Next, an example of the proposed PAL is applied to agile processes.

3.1 Structural model

Assets stored in the PAL-Wiki were identified and organised into a structure of elements. This structure is based on several software process models [35] and organisational memories in knowledge management systems [36].

The structural meta-model of the PAL-Wiki is shown in Fig. 1 as a unified modelling language (UML) package diagram. The main packages are: process users (contains various roles that fulfil knowledge management and software processes functions), process assets (contains the catalogue of processes elements and their associated assets. It also includes metadata and assets versions), assets measurement (contains reports about access to Pal-Wiki), project and context knowledge (application context of the

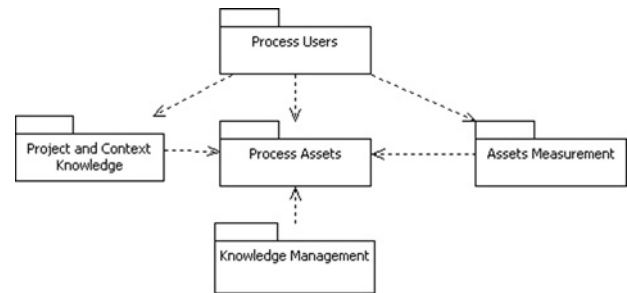


Figure 1 UML package diagram for PAL-Wiki

asset in specific projects) and knowledge management (includes elements for acquisition, organisation, distribution and utilisation of process knowledge).

An extract of the meta-model that represents this structure is presented as a UML class diagram in Fig. 2. The most important ideas underlying the model can be summarised as follows.

PAL-Wiki consists of a set of process elements: The process elements can be standard processes, life cycle models, tailoring guidelines, measurements of processes or work environment standards. The process elements are associated with many process assets that can contain formal content (structured elements such as checklists, templates etc. that implement codification strategies) or informal content (unstructured elements such as lessons learned, discussions etc. that implement personalisation strategies). The process assets may have many versions and can be instantiated and reused (asset instance) in specific projects. During the projects, there are three kind of roles: process team, responsible for producing, evaluating, editing and enhancing knowledge; development team responsible for consuming knowledge stored during the projects; and PAL administrators responsible for managing users and contents. Reports on users' access (access historial) and process assets (use history) are obtained from actions performed by users in PAL-Wiki.

3.2 Processes model

The implementation of PAL-Wiki in an organisation is a gradual process consisting of three kinds of functions based on integrative and interactive functions of knowledge management systems [37].

1. *Core functions*: First, a set of functions must be implemented with the basic knowledge management requirements during software development projects.

- *Knowledge acquisition*: This is the first function to be used in PAL-Wiki. Process knowledge is obtained with sufficient detail in order to be transferred, stored and used in the organisation. This knowledge is acquired through forms and templates for different process elements and process assets.

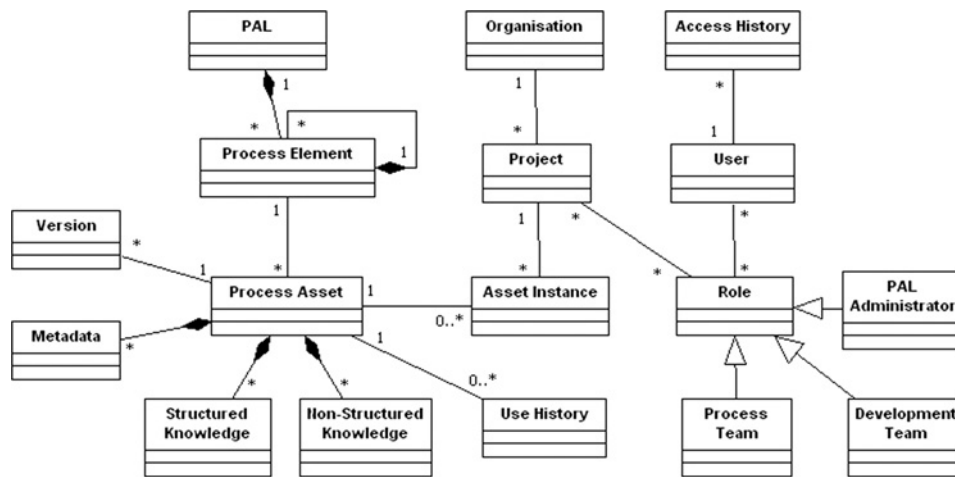


Figure 2 Extract of UML meta-model for PAL–Wiki

- *Knowledge organisation:* Next, the knowledge acquired is organised according to specific formats and categories.
 - *Knowledge distribution:* The repository should provide facilities for the effective selection of its elements through a structure with multiple types of nodes and associative connections in order to browse, view and search their contents.
 - *Knowledge utilisation:* The stored assets can be searched, downloaded and used in order to perform activities and tasks in specific projects.
 - *Knowledge reuse:* The assets adapted for specific projects can be stored as examples into the PAL–Wiki depending on the application context.
 - *Knowledge preservation:* Finally, the contents of the PAL–Wiki are updated through a collaborative improvement process via editions and feedback from users.
2. *Support functions:* These functions address activities that apply more generally to the processes in the organisation.
- *User management:* Functions for managing access policies of groups and users of PAL–Wiki.
 - *Change management:* PAL–Wiki incorporates a change control system for the contents of the process assets.
 - *Knowledge measurement:* Process improvement opportunities are identified through the collection of quantitative measures on users' access and actions.
3. *Advanced functions:* These functions address processes that are targeted towards the projects and deployment of processes.
- *Small settings knowledge:* The repository provides assets with summary and detailed contents for small and medium enterprises.

- *Projects knowledge:* The repository provides capabilities to store the work products developed from the usage of assets stored in PAL–Wiki during the projects and generating metrics about projects [15, 16].
- *Knowledge visualisation:* The repository includes knowledge maps to simplify, highlight and navigate through knowledge stored, providing a framework for organisation under topics and metadata.
- *Knowledge context:* For more effective reuse of assets, structuring knowledge first requires decontextualisation (elimination of application context) and then subsequent recontextualisation when applied to a new project.

3.3 Experience with agile processes

The PAL–Wiki meta-model, based on knowledge management, was instantiated for validation using elements of agile process. PAL–Wiki included all core and support functions. The main elements of this model are shown as a UML object diagram in Fig. 3. A brief description of the elements of the model is described below.

- *Processes:* Agile processes as user stories management, planning, development, refactoring, tracking, continuous integration and defect management [37].
- *Activities:* Each process is decomposed into a set of activities. These activities are similar in structure to processes and are broken down into tasks.
- *Tasks:* Set of actions to be undertaken to successfully perform an activity. The tasks are bottom level elements that are not decomposed.
- *Roles:* Three roles are defined: Team coordinator, responsible for co-ordinating the project; software engineers who develop the software product; and customer/user, who provides the requirements of the software product to be developed.

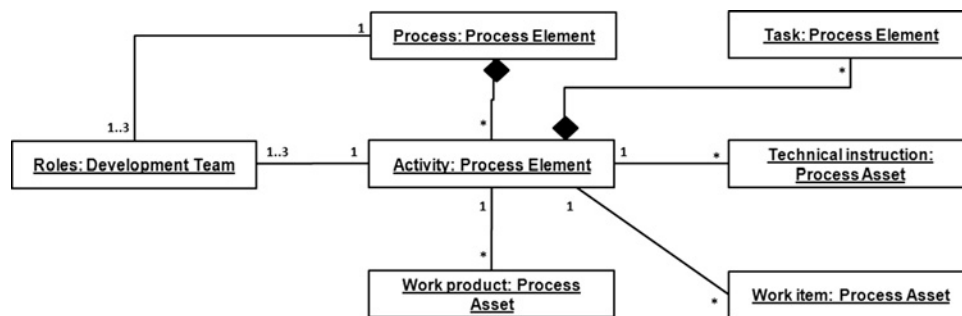


Figure 3 PAL–Wiki model oriented to agile processes

- *Work items:* Records used to track and determine the progress of an activity or task. Work items defined are: defect, user story, quality of service requirement, resource, acceptance test, unit test, class, task and automatic integration element.
- *Work products:* They are files, documents, specifications, binary files and other items that are necessary to complete the activities.
- *Technical instructions:* They provide information on carrying out activities and tasks. They can include additional resources such as slides, videos, examples and links to discussion pages.

In total, PAL–Wiki has 7 processes, 31 activities, 9 work items, 6 work products, 3 roles, 22 sets of slides, 38 videos and 30 examples. Regarding implementation issues, PAL–Wiki was built by MediaWiki tool and was installed on an Ubuntu Server. It can be reached at <http://wikisel.sel.inf.uc3m.es/dhip/>, login: guest, pass: guest01.

4 Validation description

This section identifies our research hypothesis and describes the validation phases.

4.1 Hypothesis

We formulated the following hypothesis: IF a PAL, based on knowledge management and implemented through a Wiki, is developed following the recommendations given in Section 3 THEN it

improves agile development processes learning by storing and reusing the process assets and

allows software engineers to use processes learned with a greater degree of independence.

4.2 Validation phases

Pal–Wiki was validated in a training course on agile software development processes in which junior software engineers

participated. The validation was carried out in two phases, and each phase included two stages: training and project. The timelines for each phase and stage are shown in Fig. 4.

Different groups of engineers were involved in the two phases. The engineers' experience in software development process was similar but, neither group had any experience in agile methods. Phase 1 was executed by 38 software engineers who did not use PAL–Wiki. The trainers conducted a set of tutorials for the software engineers. Phase 2 was executed by 28 software engineers, using PAL–Wiki, and tutorials were not needed.

The training stage in both phases consisted of eight practical sessions on different aspects of agile development: (a) user stories, (b) acceptance tests, (c) coding standard, (d) simple design, (e) refactoring, (f) unit tests, (g) collective ownership and continuous integration and (h) tracking. The practices were developed by teams of two engineers and the grades of each group were collected.

The project stage in both phases was addressed to modify a multimedia files management system by using the concepts of agile processes learned in the training stage. At the end of the project stage, the grades of work products were collected.

Grades corresponding to the quality of the work products were collected. In addition, statistics of access to PAL–Wiki were collected to assess the number of users, access frequencies and process assets consulted during the training and project stages in Phase 2. Two surveys were developed for training and project stages in Phase 2:

- *Training stage survey:* Once eight practice sessions were completed, an initial survey on the ease of use and benefits of using the PAL–Wiki to develop the practices was carried out. The questions in this survey are listed in Appendix 1. The survey consisted of 21 items ranging from 1 to 5 on the Likert scale. For this survey, the items at the ease of use level were (questions A1–A11); the usefulness level (questions A12–A21), 24 semantic differential items (questions A22–A25) and four open question items.
- *Project stage survey:* At the end of this stage, a second survey assessed PAL–Wiki's capabilities to provide knowledge

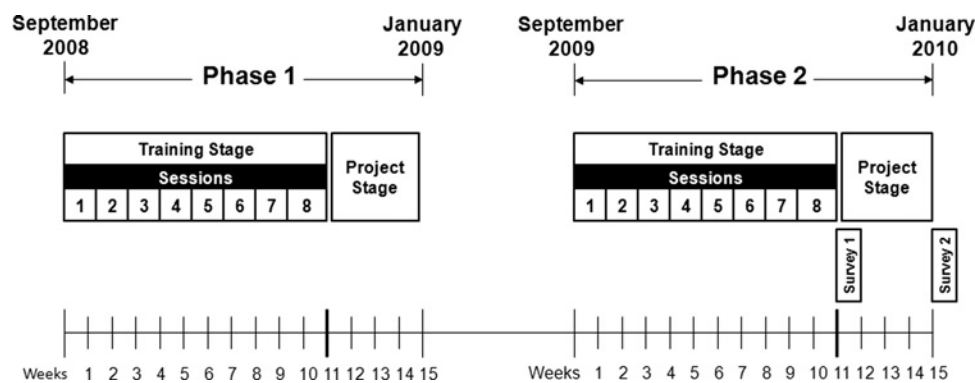


Figure 4 Timelines for validation phases and stages

during the project. The questions for this survey are listed in Appendix 2. The second survey consisted of 18 items, using the Likert scale (questions B1–B18), 10 dichotomous questions (questions B19–B28) and 13 semantic differential items with 5 rankings (questions B29–B30).

Descriptive statistics, hypothesis testing and regression and correlation analysis were used to analyse and interpret data collected.

5 Analysis of results and discussion

We are now going to discuss the results of the validation objectives of validation, starting with improving agile processes learning through storing and reusing the process assets. Next, we discuss the evidences that show PAL–Wiki allows software engineers to use processes learned with a greater degree of independence.

5.1 Objective 1: improve agile development processes learning by storing and reusing the process assets

For this objective, surveys were conducted to assess the learnability at the end of training and project stages in Phase 2. Hypothesis tests and regression and correlation analysis were applied to determine the relationships between the use of the PAL–Wiki and its benefits to perform tasks during Phase 2.

The usefulness of stored artefacts was assessed positively, specifically for examples, videos and work product because they provide a practical approach for performing tasks. Processes such as ‘refactoring’, ‘user stories management’, ‘planning’ and ‘tracking’, which included many artefacts were positively valued. In general, ease of use was considered good (3.93 on a scale of 1–5). The training survey indicates that PAL–Wiki is easy to learn and use.

Table 1 presents some statistics on access to the processes during the training stage of Phase 2 together with the learnability of each process from the survey.

A multiple regression was found ($R^2 = 0.997$) among the variables: users (U), accesses (A), time (T) and downloaded assets (D) that act as predictors of learnability (L). The equation of regression (1) obtained was

$$L = 3.58 - 0.0157 U + 0.000884 A + 0.0887 T - 0.00513 D \quad (1)$$

Equation (1) shows that accesses, duration and query and reuse of assets stored in PAL–Wiki influence learnability of software process.

The 24 semantic differential items in the Phase 2 – training stage survey related to visual presentation (question A22), ease of access and understanding the artefacts (question A23), ease of learning (question A24) and usefulness of the artefacts (questions A25) have equal distributions, using t -tests with confidence interval of 95%.

Improving software process learning is also demonstrated from a set of correlations found in survey data in the training and project stages. Table 2 presents the variables involved in the correlations found with their corresponding coefficient of correlation.

Correlation 1 shows that access to and understanding of artefacts stored are related to their usefulness. Artefacts as work items, work products, videos, slides and examples were evaluated positively according to access and understanding perceived by engineers during the training stage.

Correlations 2 and 3 show positive relationships between the training and project stages. Correlation 2 shows that access and understanding of artefacts stored during the training stage are related to the quality of the artefacts evaluated during the project stage. Correlation 3 shows that the ease of learning of the processes is related to the quality of process information during the project stage. Both correlations indicate the influence of the training stage as a learning and understanding activity to assess the quality of process information and the artefacts stored in PAL–Wiki during project development.

Table 1 Processes access statistics gathered during the training stage

Processes	Users	Access	Medium time/user, min	Assets downloaded	Learnability (1–5 scale)
user story	53	1075	11.23	149	3.93
planning	63	1688	10.68	209	3.93
development	117	2298	7.43	163	3.60
refactoring	34	725	7.78	33	4.20
integration	48	1281	8.77	175	3.86
tracking	45	811	5.33	153	3.26

Table 2 Correlations found in training and project stages

Correlation	Variable 1	Variable 2	R ²
1	access and understanding of the artefacts stored per process from the survey in Phase 2 – training stage (question A23)	usefulness of the artefacts stored per process from the survey in Phase 2 – training stage (question A25)	0.99
2	access and understanding of the artefacts stored per process from the survey in Phase 2 – training stage (question A23)	quality of artefacts per process from the survey in Phase 2 – project stage (question B30)	0.842
3	ease of learning per process from the survey of Phase 2 – training stage (question A24)	quality of process information from the survey in Phase 2 – project stage (question B29)	0.895
4	number of accesses per process from Wiki statistics	quality of the artefacts per process from the survey in Phase 2 – project stage (question B30)	0.842

Correlation 4 indicates that the greater the amount of user access to the repository, the better users perceive the quality of the artefacts to effectively carry out their activities during the training stage.

Therefore the project stage survey shows a positive assessment of the quality of process and artefacts software process learning.

5.2 Objective 2: allow software engineers to use processes learnt with a greater degree of independence

In Phase 2, software engineers worked more independently because they had learnt about the process by searching knowledge and artefacts stored in PAL–Wiki, without the trainers' tutorials.

Data on product quality levels during the project stage in both phases were collected in order to determine that an independent work using PAL–Wiki helps processes learning. The quality of each product developed is evaluated from the completeness of all its constituent parts. A value which represents the relative importance of completing the part was assigned to each part. A list of quality criteria of work products is presented in Table 3.

Table 4 presents some descriptive statistics such as number of products (N), mean (in 0–10 scale) and standard deviation (SD) of the product quality levels. There were no differences in the quality levels of products by a t -test with confidence interval of 95%. This result suggests that the quality of products developed with or without PAL–Wiki was very similar. However, the degree of software engineers' autonomy increased because there were no tutorials in Phase 2 and the engineers used PAL–Wiki to develop autonomously the software products.

The degree of the engineers' independence in finding solutions to the problems they faced improved during the training and project stages. The degree of autonomy is a consequence of the improvement in the learning process as a result of using the assets of PAL–Wiki. These findings show that PAL–Wiki improves the use of agile processes.

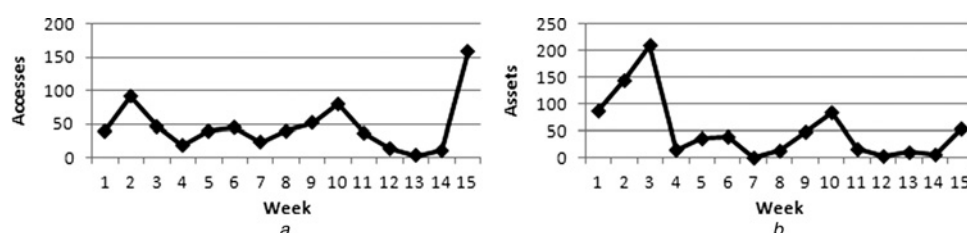
Fig. 5a shows access by week. Access was high in the first few weeks when the users did not know the PAL–Wiki structure. Then it varied according to practice deadlines. From week 12, the users began to work in the project stage and the final increase was because it was the last week to deliver the project. Fig. 5b presents a graph showing the number of assets downloaded during the course. The trend in assets downloading was similar in Fig. 5a, except the

Table 3 Quality criteria of work products

Process (value)	Criteria (value)
user story (15)	goals (1), priorities (3), size (3), efforts (3), specifications (5)
acceptance tests (10)	identification (2), detection of errors (2), purposes (2), steps (4)
planning of the iteration (13)	work capability (3), scope (1), list of tasks (1), relation to goals (1), artefacts (3), responsibilities (3), effort (3)
simple design (12)	sequence diagram (3), classes (5), classes included (1), attributes and methods (3)
refactoring (14)	coding standards (2), refactoring techniques (2), exception handling (4), controller-view communication (2), pattern 'singleton' for managing a grid (2) and a catalogue (2)
unit tests (30)	system controller (4), system of exceptions (2), pattern 'singleton' on the grid (1) and on the catalogue (1), unit tests for the goals (22)
collective ownership of code (3)	versions of the source code and unit tests (1), latest version of the artefacts (1), protected latest version of all files (1)
continuous integration (8)	integration file (1), generation of DLLs, executables and unit tests (4), outcome of automatic integration (1), files from a successful integration (1), historical record of the unit test (1)
tracking (16)	progress status (4), acceptance tests status (4), tasks status (4), accumulated value (4)
iteration wrap-up meeting (10)	meeting agenda (3), purpose (2), results (3), technical aspects (1), improvement actions (1)

Table 4 Levels of product quality

Work product	Group	N	Mean	SD	Work product	Group	N	Mean	SD
user story	1	8	10	0	unit test	1	8	4.1	1.4
	2	6	9.9	0.3		2	6	3.8	1.5
acceptance test	1	8	9.7	0.7	refactoring	1	8	9.4	1.8
	2	6	8.7	1.6		2	6	7.2	2.5
planning	1	8	8.4	1.9	tracking	1	8	8.4	2.6
	2	6	5.8	2.5		2	6	7.1	2.5
coding standards	1	8	5	5.4	iteration wrap-up meeting	1	8	7.7	2.4
	2	6	5	5.5		2	6	8.2	2.8
simple design	1	8	8	3.1	continuous integration	1	8	7.8	1.1
	2	6	7.5	1.7		2	6	8.3	1.5
collective ownership	1	8	8.4	1.3					
	2	6	8.9	1.7					

**Figure 5** Access and assets downloaded in PAL–Wiki

a Access to Wiki

b Assets downloaded

final week because they had already been downloaded in previous practices.

During Phase 2, 879 pages were viewed a total of 9028 times. The average time on site during the training stage was 8:43 minutes per user, and 22:12 minutes per user during the project stage. During the training stage, 679 pages were viewed 5598 times and during the Project stage 384 pages were viewed 3423 times.

The results show that PAL-Wiki promotes a greater degree of software engineers' independence during agile processes learning and resolves doubts about the description of the software products to be developed through storing and reusing process assets.

6 Conclusions

This paper provides two main contributions. First, a set of guidelines is defined to build a PAL, based on knowledge management, using a Wiki. To help implement the PAL, we propose a formal mechanism through the definition of the structure and functionality that a knowledge repository should have. Second, an example of the proposed PAL is applied to agile development processes. The PAL-Wiki developed is a lightweight knowledge repository orientation capturing, codifying, and disseminating knowledge about software agile processes. The core and support functions, based on knowledge management, defined in the guidelines were implemented during a course in agile development.

The PAL-Wiki developed is featured to facilitate an effective learning environment in order to help users learn an agile process, with a greater degree of independence from software engineers, to provide mechanisms for sharing knowledge, and provide a repository with useful artefacts. The results have shown that software engineers reused the stored artefacts to learn this process. The software engineers saw the benefits of PAL-Wiki. These results are consistent with the evidence that point to Wikis as scalable ways for documentation on processes and projects [23].

The results show that software engineers access, query and reuse different artefacts stored in order to learn the agile process. The repository has a positive impact on learning and using the agile process, providing useful and easily accessible support material that contributes to achieving agile practices in projects.

Software engineers reported that the PAL-Wiki is easy to learn, use and operate; the stored artefacts are useful for solving learning practices; it is a useful mean of publishing knowledge, and the reuse of assets improves the quality of the products.

The main aspect of knowledge management about software process that PAL-Wiki resolved is the development of a technology to support knowledge

management activities of agile process assets based on an integrated approach to codification and personalisation of knowledge. Other aspects of knowledge management such as support interpretation and application context, and reuse of non-structured elements are defined in the structural and functional models of the knowledge repository but they have not been implemented in PAL-Wiki. Empirical studies should investigate this in more detail.

The functions of a knowledge repository are quite general. We recommend starting with the implementation of the basic functions. The validation replication can be performed in other development processes, adapting PAL-Wiki data acquisition templates to specific processes. Moreover, the surveys conducted can be tailored to other software processes training courses.

Our findings indicate that a Wiki helps software processes learning; it improves the use of agile processes and motivates the engineers to explore concepts autonomously in order to institutionalise the knowledge about software process in organisations.

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9 Appendix 1: training phase survey

Items in Likert scale	
A1. Visual presentation	A2. Ease of browsing
A3. Structure	A4. Ease to search
A5. Contents	A6. Easy to learn
A7. Ease of use of artefacts	A8. Help available
A9. Easy to remember the structure	A10. Voluntary access
A11. Ease of use	A12. Quality of final products
A13. Practices in less time	A14. Ease for performing the practices
A15. Learning of agile processes	A16. Information for performing the practices
A17. Useful and reusable artefacts	A18. Visibility
A19. Benefits communication	A20. Content consistent with the practices
A21. Storing and publishing knowledge	

Continued

Continued

Semantic differential items	
A22. Visual presentation	A23. Ease of access
A24. Ease of learning	A25. Usefulness of the artefacts
Open questions	
A26. Which sections did you consult most?	A27. What problems did you have with it?
A28. What benefits did you obtain?	A29. What improvements do you propose?

10 Appendix 2: project phase survey

Semantic differential items	
B1. Search of information	B2. Structure
B3. GUI	B4. Interaction
B5. Control of activities	B6. Artefacts for agile process
B7. Productivity	B8. Contents
B9. Reuse of artefacts	B10. Community of users
B11. Publishing of products	B12. Repository of documents
B13. Discussions	B14. Contributions of users
B15. Awareness of processes	B16. Sharing knowledge
B17. Environment for learning of processes	B18. Using in future courses
Dichotomous questions	
Which actions did you perform in PAL–Wiki?	
B19. Watch slides	B20. Watch videos
B21. Download samples	B22. Download work products
B23. Use the search engine	B24. Read comments in discussions
B25. Participate in discussions	B26. Upload samples
B27. Add content	B28. Make small corrections
Semantic differential items	
B29. Quality of the information of the processes	B30. Quality of the information of the artefacts